

Extended exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"**Exercise: Creating a slab geometry****1. Generating an input structure for a slab**

- Coordinate files for the bulk structure should be provided, e.g. downloaded from [the Materials Project webpage](#). (To download corresponding CIF files, you have to create an account, log in and click on the atom types of interest on the periodic table. To get a starting structure for CdSe, e.g., you would choose 'Cd' and 'Se'. After clicking on search, a list of possible crystal structures will be provided from which you can choose the desired one. Doing so, the unit cell of the chosen crystal structure is visualized and by clicking on the last item at the upper right corner of the visualization window, the corresponding CIF file can be downloaded to your local machine.)
- Use VESTA to open the CIF file.
- Choose the option `Edit → lattice planes → NEW`.
- Insert the desired Miller indices, e.g. 111 to define the lattice plane that VESTA should visualize.
- After visualizing the lattice plane, choose `Edit data → Edit unit cell`. Choose the option `transform` and insert as rotation matrix the one that is required for the chosen lattice plane, e.g. for a 111 plane, the rotation matrix should be chosen as

$$\mathbf{R} = \begin{pmatrix} 0.5 & 0.5 & 1 \\ 0.5 & -0.5 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

For a 110 surface, the rotation matrix should be chosen as

$$\mathbf{R} = \begin{pmatrix} -1 & 0.5 & 1 \\ 1 & -0.5 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

For a 101 surface, the rotation matrix should be chosen as

$$\mathbf{R} = \begin{pmatrix} -1 & 0 & 2 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Click `Apply` and confirm that VESTA should "Do nothing". The unit cell should now be transformed in such a way that the lattice plane is a horizontal plane with atoms above and below.

- Remove all atoms above the lattice plane as well as all redundant atoms at the edges or facets of the unit cell (VESTA is duplicating some atoms and by keeping in mind, that the structure will be periodically replicated, all these duplicated atoms have to be removed manually).

- Export the XYZ coordinates as well as the transformed lattice parameters and copy the XYZ coordinates to the Caucluster using scp

```
scp -r CdSe111.xyz suphcXXX@caucluster.rz.uni-kiel.de:~/
```

- Switch to your \$WORK directory and create a new directory for the computation, enter the directory and move the CIF file to this new directory

```
cd $WORK
mkdir Single_point_CdSe_111_slab
cd Single_point_CdSe_111_slab
mv ~/CdSe111.xyz .
```

With `ls` or `ll` and `pwd`, you can again check that the last steps were successful.

- Copy and modify the exemplary input file `semiconductor.inp`,

```
cp /zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_general/semiconductor.inp .
```

changing the specified name of the XYZ file to be read in by CP2K.

- To generate the slab structure, a CP2K computation implying multiple unit cells (generating a larger slab by replicating the unit cell) has to be performed. To do so, the keyword `MULTIPLE_UNIT_CELL` has to be inserted in the sections `&CELL` and `&TOPOLOGY`.

```
&CELL
...
MULTIPLE_UNIT_CELL 5 3 1
&END CELL
...
&TOPOLOGY
...
MULTIPLE_UNIT_CELL 5 3 1
&END TOPOLOGY
```

- To run the computation, copy the bash script for slurm to your local directory,

```
cp /zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercises1/run_with_caucp2k.sh .
```

- Open `run_with_caucp2k.sh` with `vim` and modify the line that specifies input and output files given to CP2K according to your needs,

```
mpirun -np 32 cp2k.psmf -i XXX.inp -o cp2k.out
```

- Submit the job with

```
sbatch run_with_caucp2k.sh
```

- CP2K prints the generated coordinates in the output file under the heading "MODULE QUICKSTEP: ATOMIC COORDINATES IN ANGSTROM", listing atom, kind, element, X, Y, Z, Z(eff) and the atomic mass. Copy the columns atom, X, Y, Z of all coordinates to a new file called e.g. `CdSe111_531.xyz`. Furthermore, copy the enlarged lattice parameters from the lines starting with the prompt `CELL| Vector a`, `CELL| Vector b`, `CELL| Vector c`.

- Set up a new input file, now referring to the new coordinates `CdSe111_531.xyz` and the enlarged unit cell. Extend the c lattice vector by 10 - 20 Å of vacuum.

- Optimize the slab geometry by keeping the lowest layers of the slab fixed, specifying these atoms with the subsection `&FIXED_ATOMS` in the section `&CONSTRAINT`, a subsection of the section `&MOTION`.